

Protein binder purification in bacteria

DOI: SP000P-N

Authors	¹ Tomas Pascoa, ¹ Tina Bohstedt, ¹ Jesper S. Hansen, ¹ Huanyu Li, ¹ Sagar Raturi, ¹ Andreea Scacioc, , ¹ Abilasha Balakrishnan & ¹ David B. Sauer
Affiliations	¹ Centre for Medicines Discovery, University of Oxford. Oxford, UK

Protocol description

This protocol is used to purify the protein binders (nanobodies, sybodies, sFABs, etc) cloned into pBXNP_1xFLAG-His from periplasm of MC1061 bacteria.

Materials

Biological materials:	▪ pBXNP_1xFLAG-His is a modified E.coli expression vector for FX cloning system, N-terminal pelB signal sequence followed by the binder and C-terminal 1xFLAG and 10xHisTag (original vector Addgene #110098). ▪ MC1061 bacteria (Lucigen, cat. no. 60514-1)
Reagents:	▪ Agar, pure powder (ThermoFisher Scientific, catalog number 400400050) ▪ Miller's LB broth (MELFORD, catalog number L24045-5000.0) ▪ Carbenicillin (VWR, catalog number J61949.06) ▪ Ampicillin sodium salt (Sigma-Aldrich, catalog number A9518) ▪ HEPES (ThermoFisher Scientific, catalog number J16926-A1) ▪ NaCl (VWR, catalog number 27810.364) ▪ MgCl₂ (Sigma-Aldrich, catalog number M2670) ▪ D-glucose (Sigma-Aldrich, catalog number G7021) ▪ Sucrose (Sigma-Aldrich, catalog number S9378) ▪ Ni-NTA resin (ThermoFisher Scientific, catalog number 25215) ▪ Imidazole (Sigma-Aldrich, catalog number 56750) ▪ L-(+)-arabinose (Sigma-Aldrich, catalog number A3256)
Equipment:	▪ Gravity flow columns (VWR, catalog number 26013) ▪ 5 kDa cut-off centrifugal concentrators (Cytiva, catalog number 28932245) ▪ PD MiniTrap G-25 desalting columns (Cytiva, catalog number 28918007)

Reagents setup

Overnight (ON) medium (10mL per construct):

LB medium
2% (w/v) D-glucose
2 mM MgCl₂
100 µg/ml Ampicillin

Expression medium (100mL per construct):

LB medium
2 mM MgCl₂
100 µg/ml Ampicillin

Buffers

TES buffer
0.2 M TRIS pH 8.0
0.5 mM EDTA
0.5 M sucrose

MgCl₂ solution

5 mM MgCl₂

Phosphate buffer 1 (PB1)

50 mM NaH₂PO₄ / Na₂HPO₄ pH 7.0
1 M NaCl

Phosphate buffer 2 (PB2)

50 mM NaH₂PO₄ / Na₂HPO₄ pH 6.0
1 M NaCl

50% Ni-NTA resin

pre-equilibrated with 2x MQ and 3x PB1)

Elution buffer (EXB300)

50 mM HEPES pH 7.5
300 mM NaCl
300 mM imidazole pH 8.0

Desalting buffer

20 mM HEPES pH 7.5
200 mM NaCl

Procedure

- **Transformation**

Make a fresh transformation of purified plasmid DNA into chemically competent K12 MC1061 E. coli cells by standard bacterial transformation procedure. Plate onto agar plates containing 100 µg/mL Carbenicillin.

- **Expression**

Day 1

The overnight culture is made from 3-5 colonies from the fresh transformation plate by picking with an inoculation loop and adding this to the ON medium which is left to shake at 400 RPM at 37°C.

Day 2

1. Measure the OD₆₀₀ of the overnight starter culture.
2. Dilute the overnight culture into the 100 ml expression medium to give a starting OD₆₀₀ of 0.02-0.05.
3. Once the OD₆₀₀ reaches 0.5 (approx. 1.5-2 h), drop the temperature of the shaker to 22°C and continue to grow for 1.5 h.
4. Induce expression by adding L-(+)-arabinose to a final concentration of 0.02% (w/v).
5. Grow overnight at 22°C, 180 rpm.
6. Harvest the cells the next day by centrifugation at 4000 x g for 20 minutes.

- **Purification**

7. Resuspend cell pellet in 5 ml ice-cold TES buffer.
8. Leave cells shaking on ice (200 rpm on an orbital platform) for 1 h.
9. Add 34 ml ice-cold 5 mM MgCl₂ and keep shaking on ice for 1 h.
10. Transfer to centrifuge tubes and spin at 15 000 x g for 1 h at 4°C.
11. Transfer the supernatant to a falcon tube and add 0.5 ml 50% Ni-NTA resin (pre-equilibrated with 2x MQ and 3x PB1) to each tube.
12. Rotate at room temperature for 1 h.
13. Spin down resin at 700 x g, 5 min, deceleration 4. Remove the flow-through.
14. Resuspend in 7.5 ml PB1 (30CV). Spin down resin at 700 x g, 5 min, deceleration 4. Remove supernatant.
15. Resuspend in 5.0 ml PB2 (20CV). Spin down resin at 700 x g, 5 min, deceleration 4. Remove supernatant.
16. Transfer the resin to a small gravity column.
17. Elute with 4 x 0.25 ml EXB300.
18. Measure protein concentration.
19. Pool protein-containing fractions.
20. Load 0.5 ml of the pooled elution sample onto a PD MiniTrap G-25 desalting columns (pre-equilibrated with 4x 2 ml desalting buffer).
21. Elute with 1 ml desalting buffer (gravity protocol).
22. Run SDS-PAGE.
23. Concentrate the desalted binders using Vivaspin 2 ml 5 KDa cut-off concentrators.
24. Aliquot and flash-freeze in liquid N₂.

Additional notes

This protocol was adapted from Zimmermann *et al.* Nat Protoc. 2020¹.

Data availability

References

1. Zimmermann I, Egloff P, Hutter CAJ, Kuhn BT, Bräuer P, Newstead S, Dawson RJP, Geertsma ER, Seeger MA. Generation of synthetic nanobodies against delicate proteins. Nat Protoc. 2020 May;15(5):1707-1741. doi: 10.1038/s41596-020-0304-x. Epub 2020 Apr 8. PMID: 32269381.